Precision and Recall

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Responsible Plant Classifier

Before you start

Week 15: identify predicted-positive and actual-positive totals.

Plan time to read, investigate and explain. Keep predictions separate from observations.

Your mission

> Start with the target: Choose the denominator that answers the question. Try the recall before opening notes.

Your learning target

I can choose the **denominator** that answers the question.

I can calculate **precision** and **recall** with a declared positive class.

I can report **undefined** ratios honestly.

Remember before reading

Close your notes first. A rough first answer helps us choose the right starting point; it is not a score for your ability.

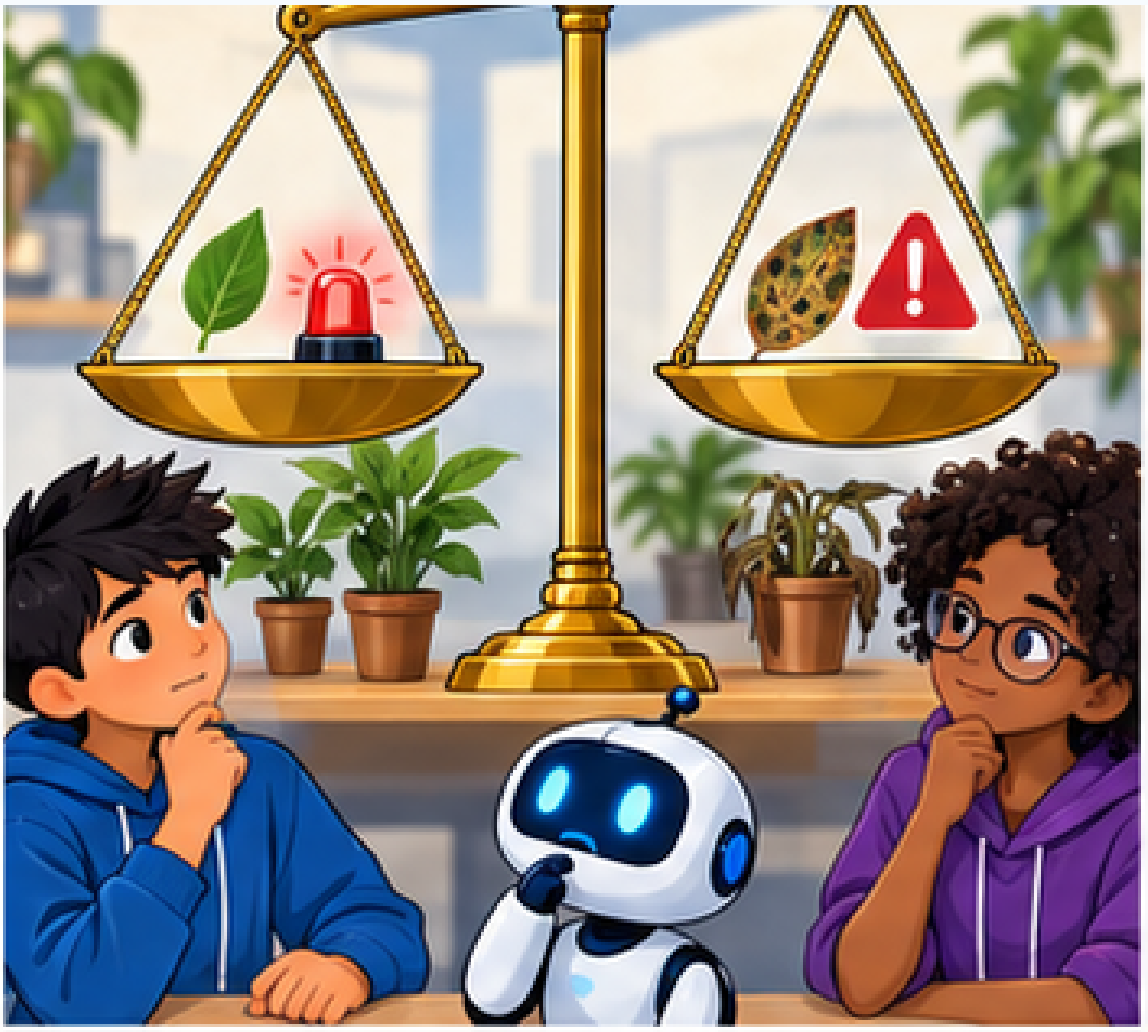
Without opening the old lesson, explain one key idea from Accuracy and Imbalance. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Scores and Calibration Intuition. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Decision Trees. Give an example and a mistake that the idea helps you avoid.

After trying, check the relevant lesson or ask your teacher. Change the part of your explanation that the **evidence** shows was wrong.

Precision measures how many positive **prediction**s are right; **recall** measures how many actual positives are found.



Your target

LEARNING TARGET TP=9,FP=3,FN=6,TN=12. Compute **precision** and **recall**. What happens to **precision** when a system predicts no positives?

Builder name and experiment date:

Readiness for the connected explanation

Why can 80 percent accuracy hide **miss**ed blocked cases?

This week's end goal

By the end, I can explain **precision** and **recall** using a traced example and a stated limitation.

STUDENT LESSON · 2 OF 18

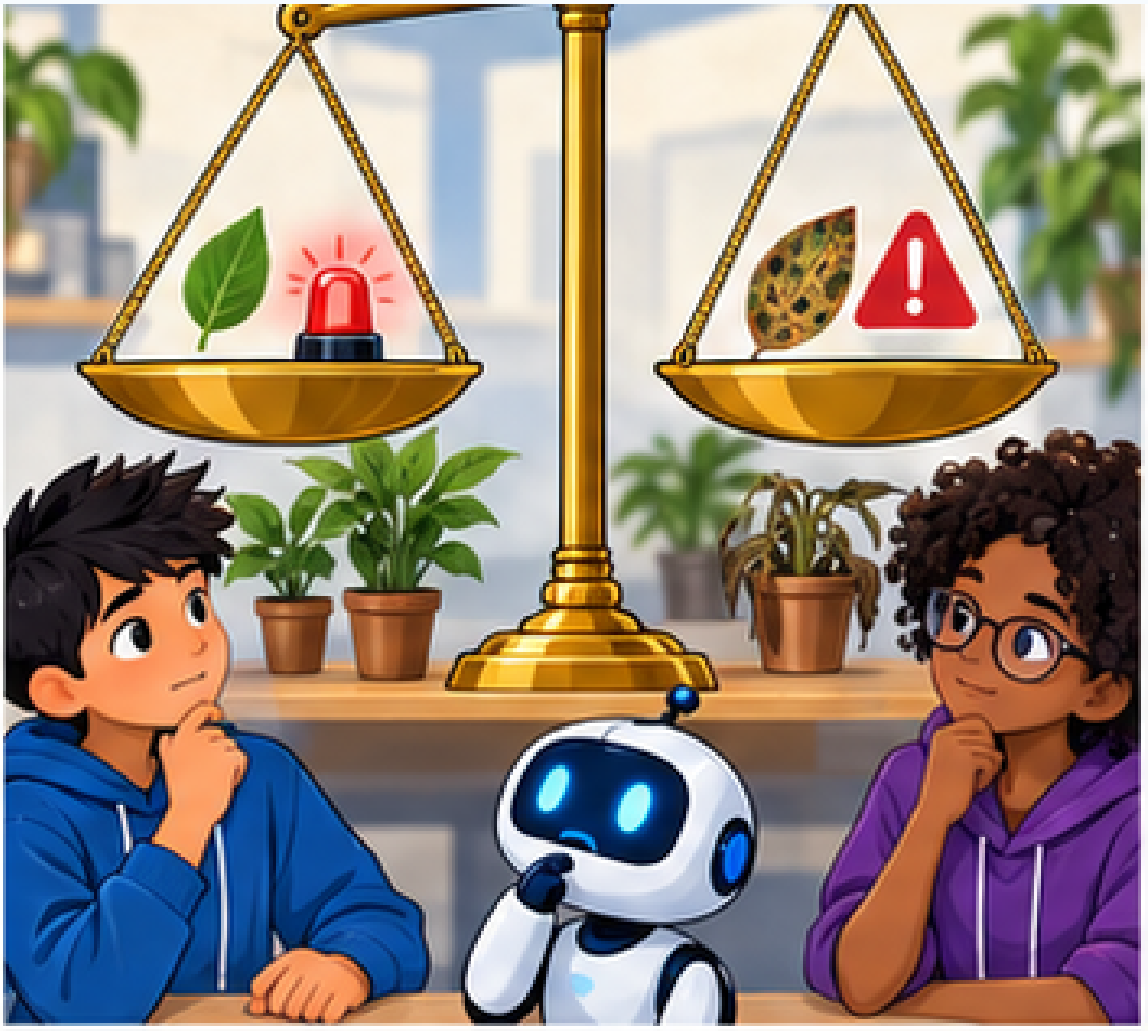
Notice the evidence

> Investigate the supplied Precision and Recall case. Separate observations, given facts and assumptions before choosing an explanation.

PICTURE ENTRY

Notice → Predict → Explain

Begin with visible **evidence**; do not explain more than the picture and information support.



Start with this puzzle

MYSTERY The team must use **precision** and **recall** to solve a model-building problem. What information do they have? What important **evidence** is still **miss**ing?

Name three visible details without interpreting them.

Predict the data, model, or evaluation decision being made.

Identify one hidden assumption that could change the result.

State what **evidence** would confirm or reject your **prediction**.

My **prediction** and supporting **evidence**:

A closer explanation

Blocked remains positive. **Precision** asks: of the cases flagged blocked, how many were actually blocked according to the reference? **Recall** asks: of all reference-blocked cases, how many were found?

Both use true positives, but they answer different questions. Keep reference rows and **prediction** columns ordered clear then blocked.

Use fictional classroom data throughout this investigation. Keep the task, units, reference labels and data split visible as you reason.

Earlier weeks supplied the model-building **process**; this week examines **precision recall** and **denominator**s.

STUDENT LESSON · 3 OF 18

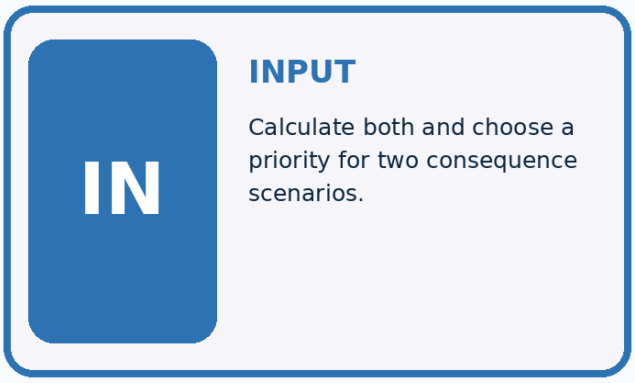
See how it works

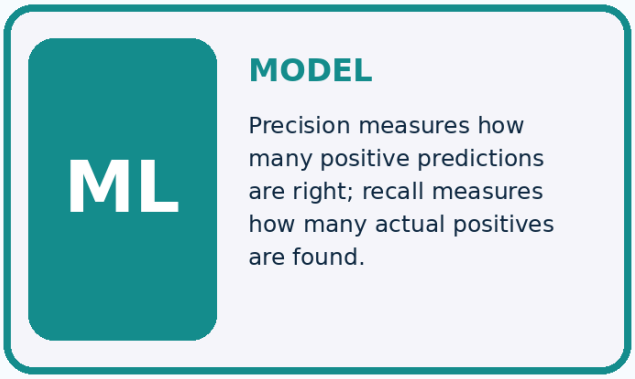
> Follow the mechanism for Precision and Recall. Connect each labelled input, deciding step and result.

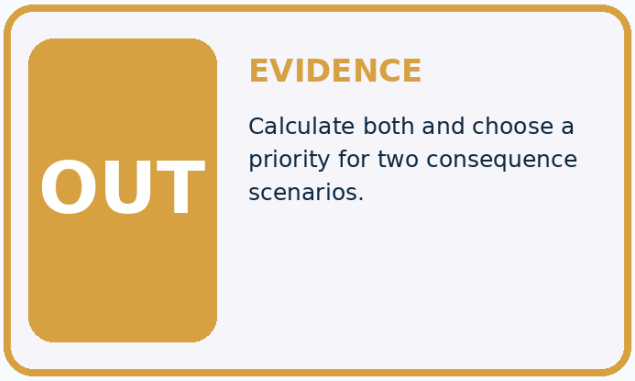
ML MECHANISM

Input → Model → Evidence

Trace the information, fitted behavior, and evaluation proof separately.







Question to explain

QUESTION Calculate both and choose a priority for two consequence scenarios.

MODEL IDEA **Precision** measures how many positive **prediction**s are right; **recall** measures how many actual positives are found.

PROOF NEEDED Use an unseen case, visible calculation, error analysis, and a stated limitation.

How the mechanism works

UNDER THE HOOD Confusion counts: TP=8, FP=2, FN=4, TN=16. **Precision** = 8 ÷ (8+2) = **80%**.

Draw a fresh data → model → **prediction** → evaluation trace:

A closer explanation

For matrix [[12, 4],[1, 3]], TN=12, FP=4, FN=1, TP=3. There are seven predicted positives and four reference positives. **Precision** is 3/7, about 42.9 percent. **Recall** is 3/4=75 percent. Accuracy is 15/20=75 percent.

A correct **denominator** matters more than memorizing a percentage.

```
tn, fp, fn, tp = 12, 4, 1, 3
precision = tp / (tp + fp) if tp + fp else None
recall = tp / (tp + fn) if tp + fn else None
accuracy = (tn + tp) / (tn + fp + fn + tp)
print(precision, recall, accuracy)
# None means unavailable, not zero or perfect.
```

Trace the calculation on paper before running this optional Python check. Use the stated fictional **input**s and compare intermediate values. The code checks this example; your independent workbook case uses different **evidence**.

Connect the picture and the explanation

Point to the **input**, the deciding step and the result in this week's visual. Explain the same step in ordinary words. A picture helps when you can say what each part represents.

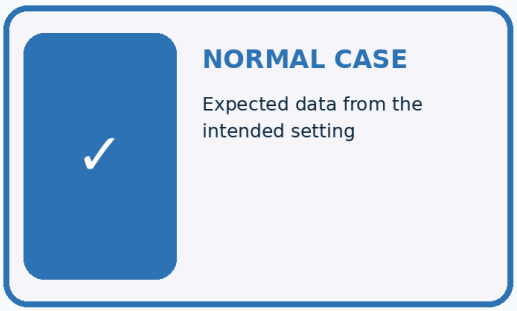
Compare the cases

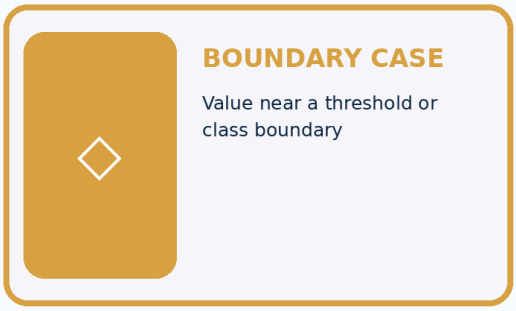
> Compare the supplied cases for Precision and Recall. Identify the detail that changes the conclusion.

EVALUATION DESIGN

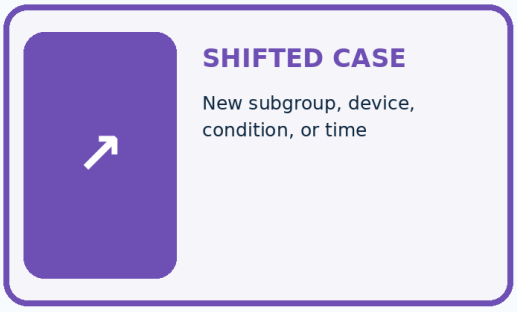
Four Tests, Four Kinds of Evidence

A system is not understood until it survives more than one comfortable example.









NORMAL Expected data and expected behavior.

Keep this boundary clear

BOUNDARY Value close to a threshold, ambiguous label, or uncertain **prediction**.

MISSING Necessary field, label, source, or measurement is unavailable.

SHIFTED New lighting, species, device, subgroup, or time condition.

RELEASE RULE Passing one case never proves generalization. Predefine expected behavior for all four categories.

Which case most threatens today's claim, and why?

Change one thing in the pictured case

Change one thing

Change: Correct two false negatives into true positives.

Predicted result

Prediction: **Precision** becomes8/10=**80%**; **recall** becomes8/10=**80%**.

Why the result follows

Reason: TP becomes8 and FN 2; FP remains2, so the two **denominator**s happen to equal10.

STUDENT LESSON · 5 OF 18

Words that unlock the idea

> Use the vocabulary of Precision and Recall to explain a concrete decision, rather than only memorising definitions.

VOCABULARY

Words You Can Calculate and Defend

A definition matters when it lets you interpret a new dataset, model, or result.

1

PRECISION

TP divided by TP plus FP

2

RECALL

TP divided by TP plus FN

3

FALSE ALARM

false-positive prediction

4

MISS

false-negative prediction

Term	Precise meaning	My ML example
precision	TP divided by TP plus FP	
recall	TP divided by TP plus FN	
false alarm	false-positive prediction	
miss	false-negative prediction	

MISCONCEPTION TO REPAIR **Precision** and **recall** are two names for the same accuracy calculation.

Repair the claim using a definition, calculation, or test record:

Words used in the closer explanation

Word	Meaning
Precision	True positives divided by all predicted positives
Recall	True positives divided by all reference positives
Denominator	The group counted in the bottom of a fraction
Undefined	A ratio with no valid value because its denominator is zero

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STUDENT LESSON · 6 OF 18

Follow a worked example

> Follow the worked trace for Precision and Recall, predict a step, then compare with its explanation.

WORKED EVIDENCE

Reason Step by Step

Predict first. Then check each step. A correct answer without visible **evidence** cannot be audited.

WEEK 17 • WORKED EVIDENCE

1. Confusion counts: TP=8, FP=2, FN=4, TN=16.
2. Precision = 8 ÷ (8+2) = 80%.
3. Recall = 8 ÷ (8+4) ≈ 67%.
4. Choose threshold and human review by comparing the cost of two false alarms with four misses.

Step 1

STEP 1 Confusion counts: TP=8, FP=2, FN=4, TN=16.

Step 2

STEP 2 **Precision** = 8 ÷ (8+2) = **80%**.

Step 3

STEP 3 **Recall** = 8 ÷ (8+4) ≈ 67%.

Step 4

STEP 4 Choose threshold and human review by comparing the cost of two **false alarm**s with four **miss**es.

CHECK Verify the **denominator**, unit, split, class definition, and data version before interpreting a score.

Recalculate, retrace, or restate the **evidence** independently:

A closer explanation

Precision recall and denominators at a glance

PART	EVIDENCE OR ACTION	INTERPRETATION
Precision	3 correct flags	7 total flags
Recall	3 found blocked cases	4 reference blocked
Accuracy	15 correct labels	20 total records

Draw a box around the predicted-blocked column for precision and around the reference-blocked row for recall. Their overlap is TP. A model can flag nearly everything and find most blocked cases while producing many false alarms. Conversely, a model can flag very few cases accurately while missing many positives.

Calculate precision, recall and accuracy for the worked matrix. Explain the steps or evidence behind your response.

Learn from the worked case

Cover the result before checking it. Say what is given, choose the procedure, and follow one step at a time. Then uncover the result and explain your first mismatch. Ask why a step is allowed, not only what number it produces.

A pictorial worked case: Precision and Recall

Put **precision** and **recall denominator**s on the picture instead of memorising names.

The facts we will use

Supplied facts

Positive means spotted. TP=6, FP=2, FN=4, TN=8 in twenty fictional leaves.

Supplied facts

All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

AI ACADEMY / AI-4/17

Keep precision and recall denominators distinct

	predicted spotted	predicted healthy
actual spotted	6	4
actual healthy	2	8

Read a cell using both its row and column labels.

Follow the labelled picture with the matching explanation. Open the vector diagram

AI ACADEMY / AI-4/17

Precision and Recall

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Predicted positives

Eight leaves were flagged: precision=6/(6+2)=75%.





2. Actual positives

Ten leaves were spotted: recall=6/(6+4)=60%.





3. Keep goals separate

Precision asks how reliable flags are; recall asks how many spotted leaves were caught.

CHANGE ONE THING

Correct two false negatives into true positives.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. Predicted positives	Eight leaves were flagged: precision=6/(6+2)=75%.
2. Actual positives	Ten leaves were spotted: recall=6/(6+4)=60%.
3. Keep goals separate	Precision asks how reliable flags are; recall asks how many spotted leaves were caught.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Why this result follows

Why the result follows

Read the picture in order. **Precision** asks how reliable flags are; **recall** asks how many spotted leaves were caught. The arrows show which recorded **input** or intermediate result each decision uses.

Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

What this example does not establish

Keep the stated limit

If a relevant **denominator** is zero, report that metric as **undefined** under the chosen convention rather than silently inventing success.

Keep the stated limit

Read the labelled picture: Were the original **precision** and **recall denominator**s the same? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

Keep the stated limit

Change one thing: Correct two false negatives into true positives. Predict the result before checking the explanation. What remains the same?

STUDENT LESSON · 7 OF 18

Find and repair a mistake

> Find the unsupported step in the Precision and Recall claim and explain a repair using the supplied evidence.

Find the first error

ERROR DETECTIVE

Compare → Diagnose → Repair

Do not patch the answer. Find the earliest unsupported assumption, data problem, or experiment error.

!

COMPARE • DIAGNOSE • REPAIR

Tempting claim: Precision and recall are two names for the same accuracy calculation. Inspect the dataset, split, calculation, error type, subgroup, or experiment record; identify one testable cause; change one factor; rerun the same evidence.

Evidence record	Expected	Actual	Likely cause / next test
normal case	passes		
boundary case	defined behavior		
missing case	safe fallback		
shifted case	measured limit		

CLAIM TO CHALLENGE **Precision** and **recall** are two names for the same accuracy calculation.

Diagnose one cause and name the cleanest next experiment:

A closer explanation

Ratios describe this evaluation set, not a guarantee for an individual case. Library zero-division policies differ; document yours. Do not select a metric only after seeing which one makes a model look best.

> **A team reports recall 100 percent but hides precision and counts. What is missing?**

STUDENT LESSON · 8 OF 18

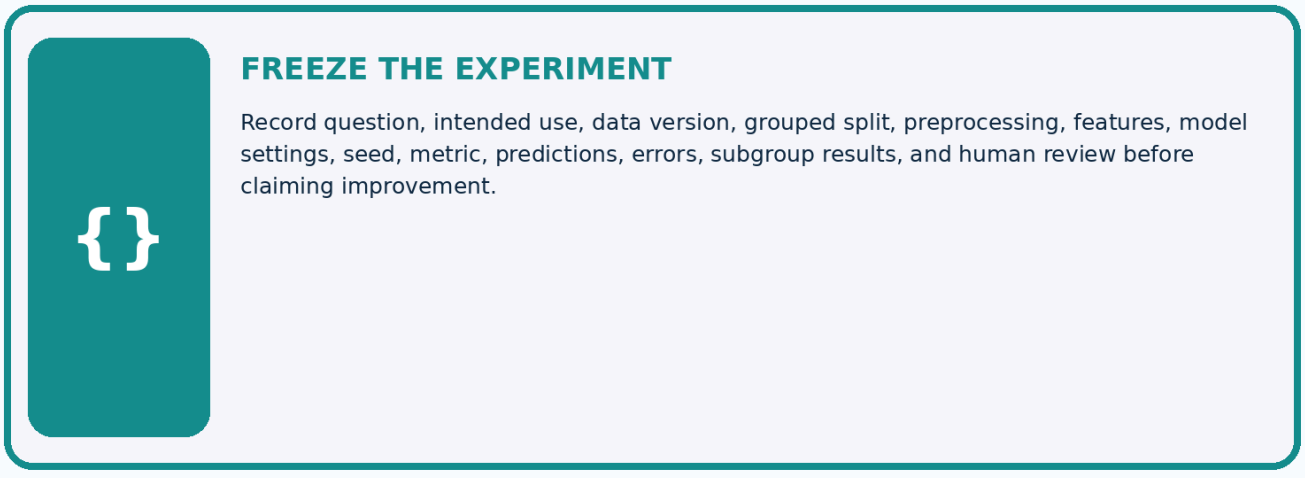
Practise together

> Practise Precision and Recall with support, then explain the deciding step in your own words.

ML STUDIO

Specify Before You Run

A trustworthy result can be reproduced from the recorded data, code, settings, and evaluation rule.



Experiment record

> **Question to explain**
>
> QUESTION + intended use + success threshold

DATA version + grouped split + leakage check

CODE/pre**process**ing + features + model + settings + seed

METRICS + positive class + units + **denominator**s

PREDICTIONS + errors + subgroup slices + limitations

HUMAN reviewer + release, revise, limit, or stop decision

Write the minimum record needed to reproduce today's result:

A closer explanation

With [[16, 0],[4, 0]], there are no predicted positives. **Precision** is **undefined** because its **denominator** is zero; **recall** is 0/4=0. With [[16, 4],[0, 0]], **recall** is **undefined** because no reference positives were evaluated.

Record counts and the **undefined** result instead of calling it perfect.

Practise with less help: Calculate both ratios

Use workbook W2 for the connected example "**Precision recall** and **denominator**s". First fill the **input** and rule boxes. Try the middle steps before asking for a hint. After a hint, try the next step yourself.

Plan	Do	Check
What is given? Which rule or procedure applies?	Record each intermediate step.	Does the result follow? What remains unknown?

Explain your trace to a partner. Your partner points to one step to check. Swap roles on the next case. Each person writes their own explanation.

Check your reasoning

Use the worked case for **Precision recall** and **denominator**s. Proposed explanation: “A final answer is enough”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Answer on your own before discussion. If you are unsure, say which step you need help with.

Explain the picture

Were the original **precision** and **recall denominator**s the same? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

> **My explanation using a pictured step and the given facts**

This is supported practice. Keep the separately named independent case for an attempt before feedback.

STUDENT LESSON · 9 OF 18

Run the investigation

> Investigate Precision and Recall. Record the prediction, actual observation or labelled paper trace, explanation and next test separately.

EVIDENCE LAB

Predict → Run → Calculate → Interpret

Keep the test conditions fixed, record every result, and change only one planned factor.

LAB

PREDICT → RUN → CALCULATE → INTERPRET

Metric Tradeoff Lab: compute both metrics for two thresholds, map errors to consequences, and defend a choice.

1. Write the hypothesis and expected **evidence** before running.

2. Metric Tradeoff Lab: compute both metrics for two thresholds, map errors to consequences, and defend a choice.

3. Run one normal, one boundary, one **miss**ing, and one shifted case.

4. Calculate the metric from raw counts or row-level errors.

5. Interpret what changed, what did not, and what remains unsafe or unknown.

Case · Prediction · Actual · Calculation / evidence · Meaning / next test

> **Case:** Normal

Prediction: _____

Actual: _____

Calculation / evidence: _____

Meaning / next test: _____

Case: Boundary

Prediction: _____

Actual: _____

Calculation / evidence: _____

Meaning / next test: _____

Case: Missing

Prediction: _____

Actual: _____

Calculation / evidence: _____

Meaning / next test: _____

Case: Shifted

Prediction: _____

Actual: _____

Calculation / evidence: _____

Meaning / next test: _____

Plan → try → check

Before trying: what do I expect, and why? While trying: which step could first go wrong? Afterwards: what did I actually observe, and what should change? Keep “not run” if you only traced the procedure.

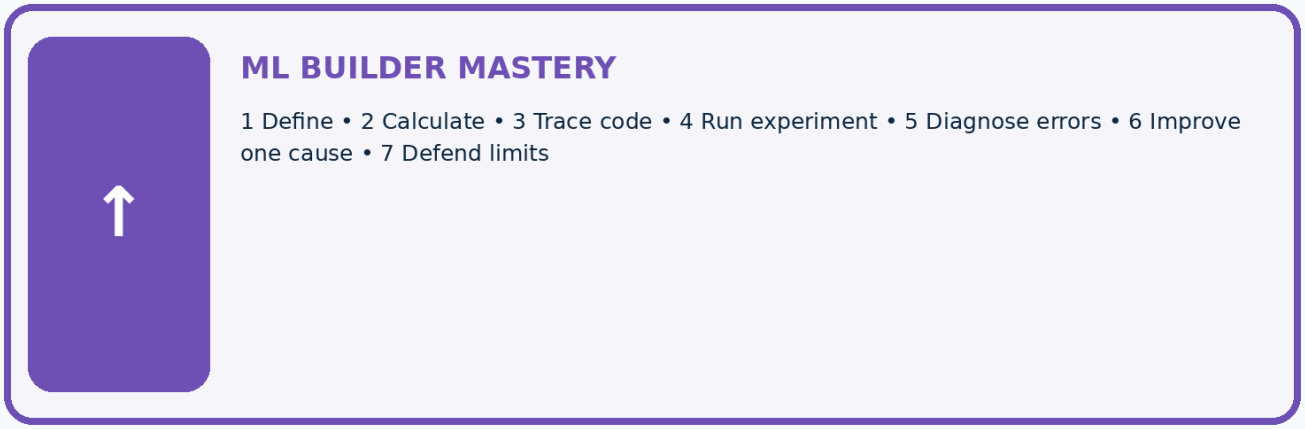
Build your understanding

> Move from recall to tracing, diagnosis and transfer in Precision and Recall. Explain why each step is justified.

MASTERY

From Knowing the Word to Owning the Idea

Move upward only when you can produce **evidence** without copying the example.



DEFINE Use all four terms precisely: **precision**, **recall**, **false alarm**, **miss**.

CALCULATE Reproduce the worked **evidence** with units and **denominator**.

TRACE Explain the data, code or decision path.

DIAGNOSE **Precision** and **recall** are two names for the same accuracy calculation.

TRANSFER Solve a fresh dataset or model case.

IMPROVE Change one cause and rerun fixed **evidence**.

DEFEND State intended use, metric, limitation, and human responsibility.

MASTERY RULE A coached calculation becomes mastery only after a different case is completed independently.

STUDENT LESSON · 11 OF 18

Connect your project

> Apply this week's idea to Responsible Plant Classifier. Keep an annotated decision and the evidence supporting it.

CUMULATIVE PROJECT

Responsible Plant Classifier

Every week adds one reviewable artifact to the same **evidence**-based capstone.

RESPONSIBLE PLANT CLASSIFIER

Set a Plant Classifier review threshold and explain which error type it prioritizes and who makes the final decision.

THIS WEEK'S CHECKPOINT Set a Plant Classifier review threshold and explain which error type it prioritizes and who makes the final decision.

Which requirement or risk does this artifact address?

Which dataset, split, code, and metric version does it use?

What normal and shifted cases were tested?

What remains uncertain or weak?

What decision still belongs to a teacher or qualified human?

RESPONSIBLE RELEASE Never use this classroom prototype for medical, pesticide, agricultural, or other consequential decisions.

My **evidence**-linked project decision:

Evidence for your project

For your Responsible Plant Classifier, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

STUDENT LESSON · 12 OF 18

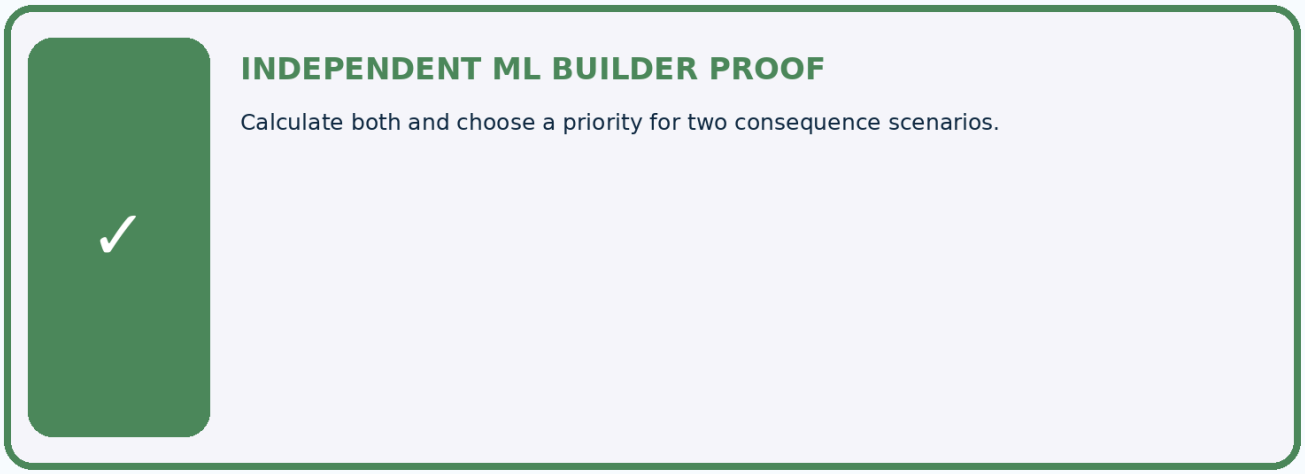
Show what you know

> Use a genuinely new case for independent evidence. Previously discussed examples remain practice; use the separately named independent case before feedback.

INDEPENDENT PROOF

Explain • Calculate • Transfer • Defend

Complete this with a fresh case after guided practice and compare it with the mastery criteria.



What success looks like

REQUIRED PROOF Calculate both and choose a priority for two consequence scenarios.

- ☐ Define the task, feature/target roles, and intended-use boundary.
- ☐ Show the calculation, code trace, or experiment record.
- ☐ Test a normal case and one revealing boundary, **miss**ing, or shifted case.
- ☐ Diagnose one failure without changing several factors at once.
- ☐ Name a limitation, fairness concern, and the human-review action.

Independent **evidence** and defense:

My level: ☐ Not yet ☐ With one hint ☐ Independently ☐ I can teach and defend it

Connect to the next lesson

NEXT BRIDGE Carry the experiment **evidence** into: Error Analysis.

Study a complete case

> Revisit the complete Precision and Recall case for supported practice. Repeated examples do not become fresh assessments.

Week 17 Worked case

Use the supplied details to practise the weekly concept before attempting the independent question. This case extends the illustrated lesson.

The situation

Positive is spotted: TP=8, FP=2, FN=4.

The question

Compute **precision** and **recall**.

Worked explanation

Precision=8/10=**80%**; **recall**=8/12 about 66.7%. **Denominator**s are predicted positives and actual positives respectively.

Explain it in your own words

> Which detail supports the answer, and why does it matter?

Write or draw your reasoning and evidence.

Trace the supplied case visually

Precision and recall use different denominators

Evidence or stage	Value or result
Precision, predicted positives	8/10
Recall, actual positives	8/12

Positive is spotted: TP=8, FP=2, FN=4.

Compute **precision** and **recall**.

Precision=8/10=**80%**; **recall**=8/12 about 66.7%. **Denominator**s are predicted positives and actual positives respectively.

STUDENT LESSON · 14 OF 18

Try a new case independently

> Attempt the reserved new case before feedback. Record support; the earlier examples below remain additional practice.

Independent case: Apply independently

Use [[8, 2],[3, 6]]. Give TP, FP, FN, **precision**, **recall** and accuracy.

> **My independent explanation, trace and limits**

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

Week 17 Independent application

Close the worked explanation. Apply the idea to this different question without copying.

With disease positive, TP=6, FP=2 and FN=4. Calculate **precision** and **recall**, showing both **denominator**s.

My answer, with a trace, calculation or supporting **evidence**

> **Write or draw your reasoning and evidence.**

Create and test

Change one relevant detail. Predict the result before testing. Include a boundary or **miss**ing-information case when it helps reveal a limit.

My changed **input** and expected result

> **Write or draw your reasoning and evidence.**

Connect to your project

Set a Plant Classifier review threshold and explain which error type it prioritizes and who makes the final decision.

The **evidence** I will keep

Write or draw your reasoning and evidence.

Your teacher has the independent answer and scoring criteria. Complete the matching workbook week to keep your practice **evidence** together.

Transfer practice from the original programme

TP=9,FP=3,FN=6,TN=12. Compute **precision** and **recall**. What happens to **precision** when a system predicts no positives?

Explain the mechanism, evidence and one limit.

Use feedback to improve

Attempt the independent case before hints. Record whether you worked alone, used a hint or followed a model. After feedback, fix the first unsupported step and explain why it changed.

A later check uses a changed case, not a copied answer.

Your lesson route

Use this map to navigate Precision and Recall, connect the goals to tasks and keep one evidence record for each attempt.

Start with this question: Put **precision** and **recall denominator**s on the picture instead of memorising names.

By the end, I can	How I will show it
Choose the denominator that answers the question	Explain the deciding step, show a trace or test, and state one limit.
Calculate precision and recall with a declared positive class	Explain the deciding step, show a trace or test, and state one limit.
Report undefined ratios honestly	Explain the deciding step, show a trace or test, and state one limit.

Remember → watch and explain → try with help → investigate → try independently → keep project **evidence**. Your teacher may spread this lesson across several classes.

Keep your first **prediction**. Compare it with what the **evidence** shows and explain any change.

STUDENT LESSON · 16 OF 18

Finish and return

> Explain Precision and Recall without notes, identify a remaining question and plan a later check with a changed case.

Close your notes. Explain this goal in your own words: Choose the **denominator** that answers the question

My deciding step, evidence and one remaining question

If you are unsure, return to the worked picture and explain one arrow. If you are ready, change an **input** or assumption and justify your **prediction**.

Save your workbook **evidence**. At the next review, try a different case before opening your old answer.

STUDENT LESSON · 17 OF 18

Supported practice, repair and portfolio

> Attempt supported questions before using hints or feedback. Review a missing building block, then keep the completed work.

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Put **precision** and **recall denominator**s on the picture instead of memorising names.

Practice 1: explain the deciding evidence

Which labelled step matches this detail? **Precision** asks how reliable flags are; **recall** asks how many spotted leaves were caught.

1. Predicted positives

2. Actual positives

3. Keep goals separate

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Feedback · compare after trying

After your attempt, compare with the supported explanation: Keep goals separate **Precision** asks how reliable flags are; **recall** asks how many spotted leaves were caught. Read the picture in order.

Precision asks how reliable flags are; **recall** asks how many spotted leaves were caught. The arrows show which recorded **input** or intermediate result each decision uses.

Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

Practice 2: explain the deciding evidence

Changed case · predict first

Changed-case practice: Correct two false negatives into true positives. Which conclusion is supported by the supplied facts?

1. We can decide the result without examining the changed **input** or the procedure.

2. This one example guarantees correct results for every future **input**.

3. **Precision** becomes8/10=**80%**; **recall** becomes8/10=**80%**.

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Feedback · compare after trying

After your attempt, compare with the supported explanation: **Precision** becomes8/10=**80%**; **recall** becomes8/10=**80%**. TP becomes8 and FN 2; FP remains2, so the two **denominator**s happen to equal10.

Repair a missing building block

Review Accuracy and Imbalance (ai-4/16). Goal: Explain why overall accuracy can hide failure on a less common class. Short practice: Why can the 90% baseline still be unsuitable for a detection task?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Review Scores and Calibration Intuition (ai-4/14). Goal: Interpret calibration using groups of forecasts. Short practice: Does a calibrated 0.8 score mean every such leaf is spotted?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Review Decision Trees (ai-4/11). Goal: Fit a root question and leaf **prediction**s from **training** examples. Short practice: Which **input** distinguishes the original and changed tree?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Predict, run and debug

The browser lab lets you edit the supported Python examples already included in this lesson. Predict first, run, inspect the actual **output** or error, change one **input**, then run again.

Keep code, **prediction**s, actual results and a reasoned revision together. On paper, label your result as a trace rather than an execution.

Keep your completed work

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

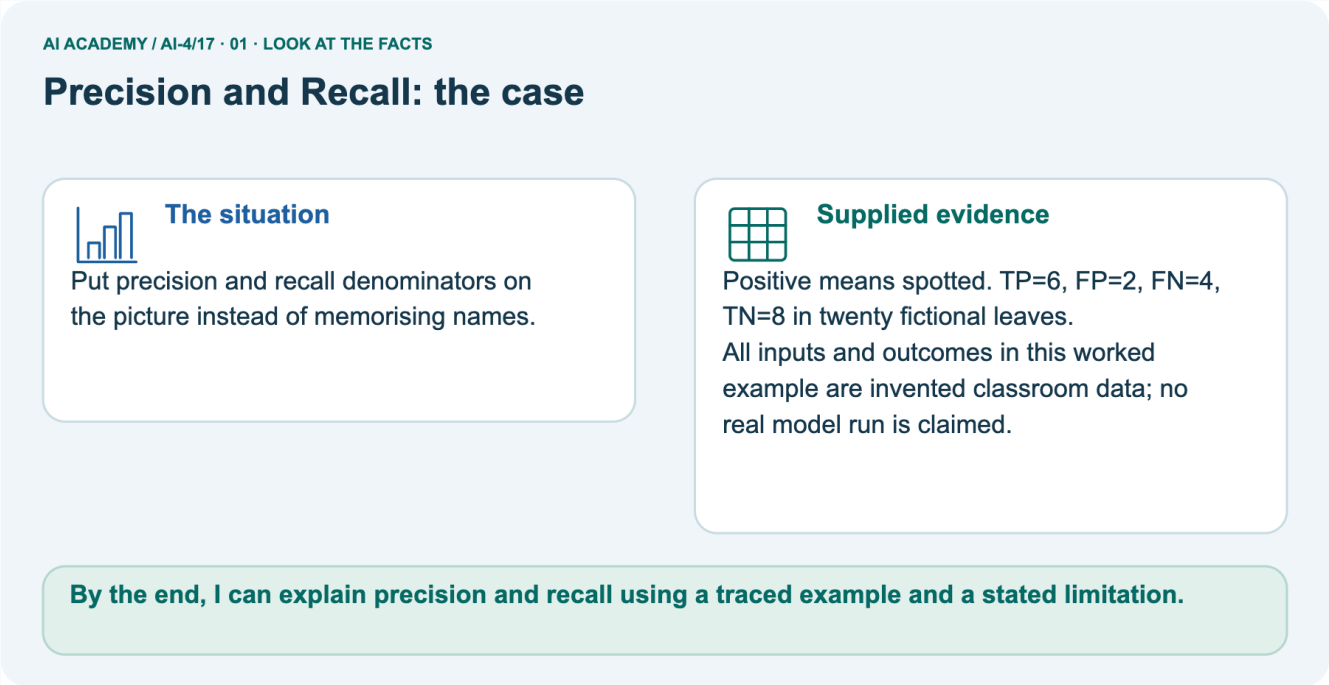
Drafts stay in this browser when storage is available; clear drafts on a shared device.

STUDENT LESSON · 18 OF 18

Visual concept journey

> Explain each Precision and Recall graphic using its labels, deciding evidence and relevant limitation.

01. Precision and Recall: the case



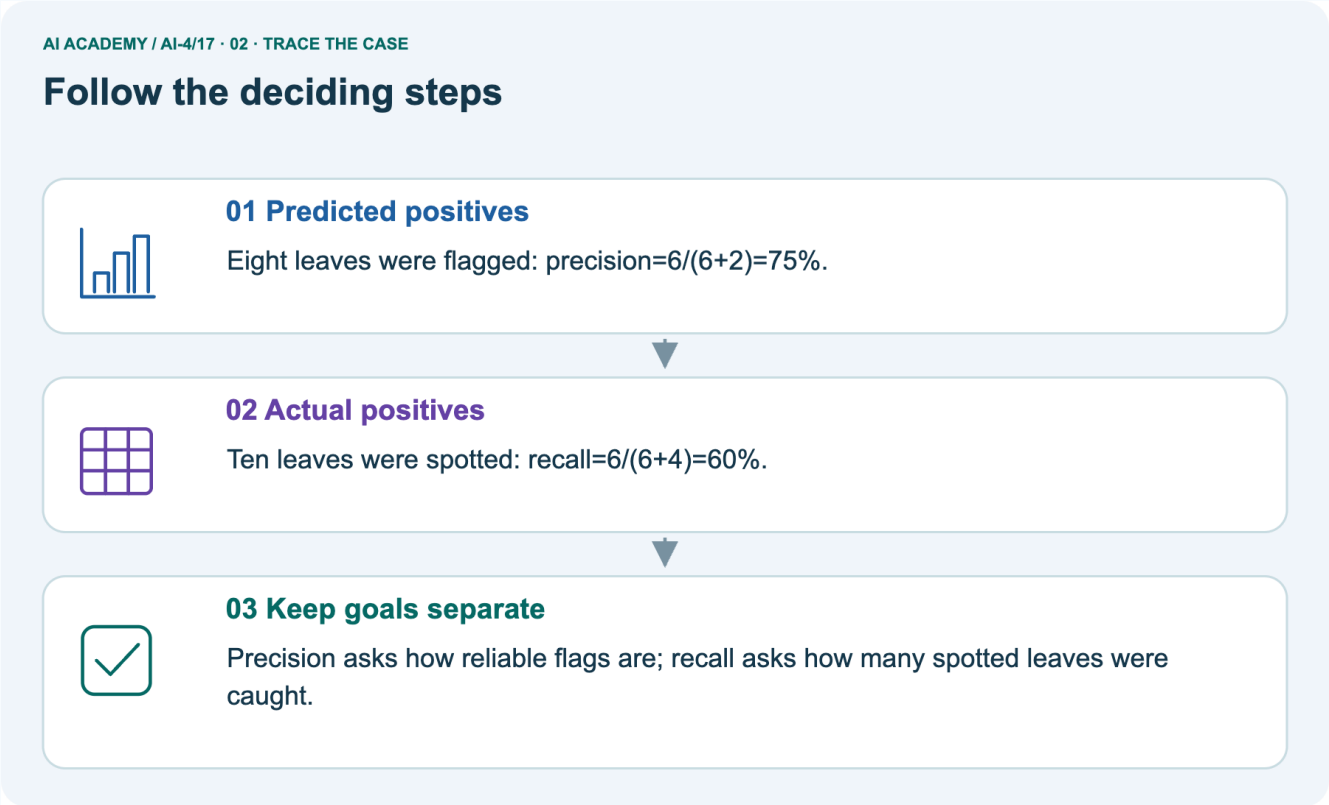
Follow the labelled picture with the matching explanation. Open the vector diagram

Put **precision** and **recall denominator**s on the picture instead of memorising names. Supplied facts: Positive means spotted. TP=6, FP=2, FN=4, TN=8 in twenty fictional leaves.

All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

> Point to the given fact that matters most. Explain why it matters.

02. Follow the deciding steps

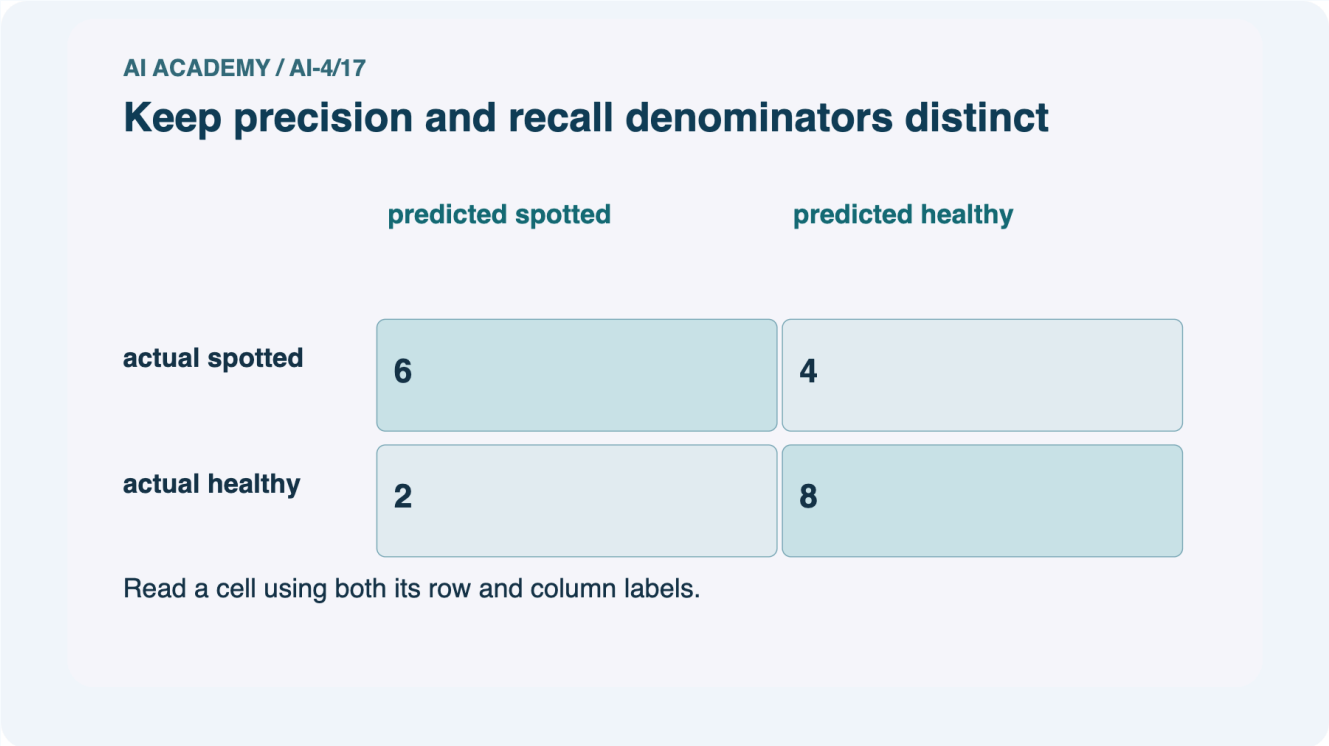


Follow the labelled picture with the matching explanation. Open the vector diagram

Predicted positives: Eight leaves were flagged: **precision**=6/(6+2)=75%. → Actual positives: Ten leaves were spotted: **recall**=6/(6+4)=**60%**. → Keep goals separate: **Precision** asks how reliable flags are; **recall** asks how many spotted leaves were caught.

> Point to each numbered stage and explain which recorded **input** it uses.

03. Keep precision and recall denominators distinct

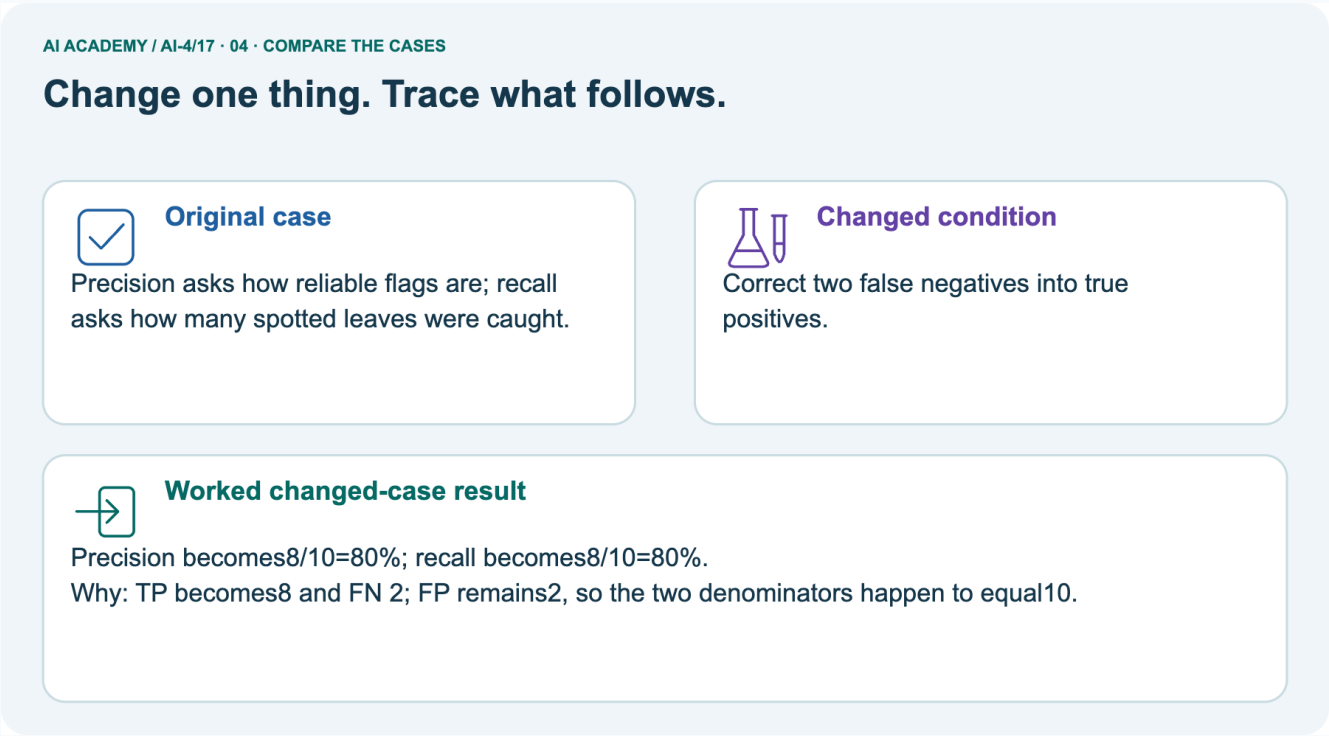


Follow the labelled picture with the matching explanation. Open the vector diagram

Keep **precision** and **recall denominator**s distinct. actual spotted: predicted spotted = 6, predicted healthy = 4. actual healthy: predicted spotted = 2, predicted healthy = 8.

> Read the labels and values. Explain the relationship the model shows.

04. Change one thing. Trace what follows.



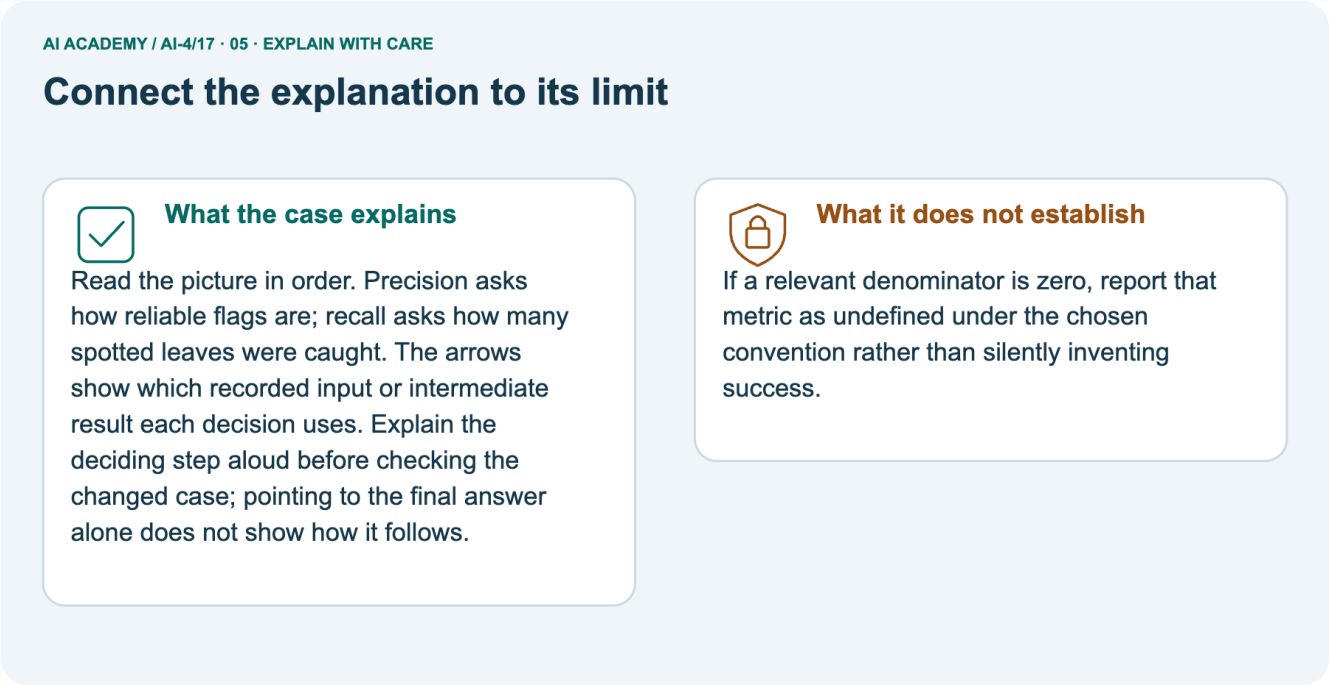
Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: **Precision** asks how reliable flags are; **recall** asks how many spotted leaves were caught. Changed condition: Correct two false negatives into true positives.

Expected result: **Precision** becomes8/10=**80%**; **recall** becomes8/10=**80%**. Reason: TP becomes8 and FN 2; FP remains2, so the two **denominator**s happen to equal10.

> Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

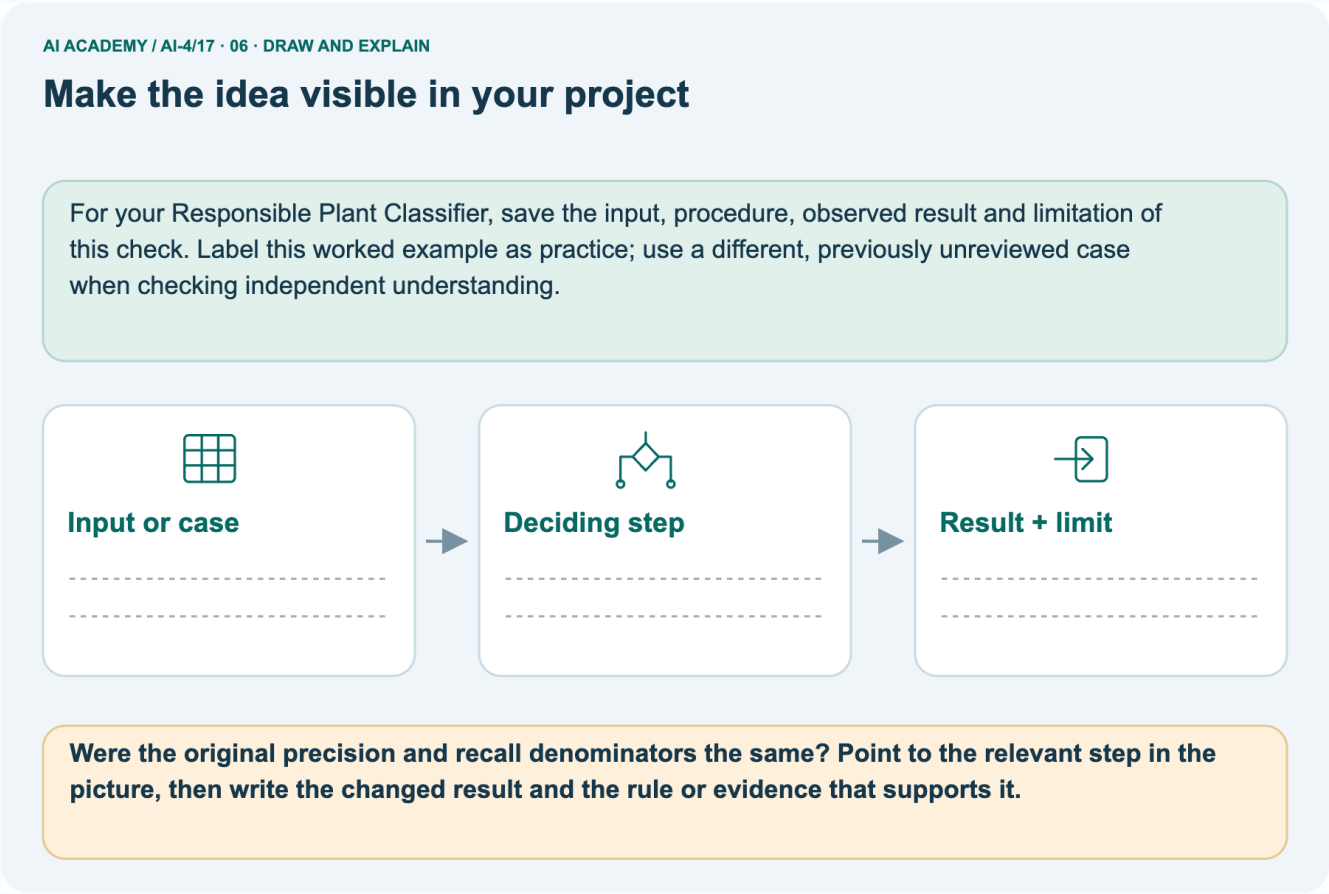
Read the picture in order. **Precision** asks how reliable flags are; **recall** asks how many spotted leaves were caught. The arrows show which recorded **input** or intermediate result each decision uses.

Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

Limit: If a relevant **denominator** is zero, report that metric as **undefined** under the chosen convention rather than silently inventing success.

> Explain the case, then point to a limit on the claim.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For your Responsible Plant Classifier, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding. Draw your **input**, deciding step, result and limit.

Supported question: Were the original **precision** and **recall denominator**s the same? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

> Draw your project trace and justify the deciding step using **evidence**.

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.